

Mind the Gap: A Non-Adversarial Framework for Unsupervised Translation

Vitor Cabral de Oliveira¹

¹Centro de Informática – Universidade Federal de Pernambuco (UFPE)
Recife – PE – Brazil

vnco@cin.ufpe.br

Abstract. *This paper introduces a non-adversarial method for unsupervised translation between heterogeneous embedding spaces. Prevailing approaches, such as vec2vec [Jha et al. 2025], often rely on adversarial training based on CycleGAN [Zhu et al. 2020], which can suffer from instability [Saxena and Cao 2023]. We move beyond this paradigm by proposing a novel architecture composed of two distinct autoencoders coupled via a shared latent space. This latent space is trained to function as an interlingua, enabling robust translation without requiring paired data. By replacing the adversarial objective with a combination of reconstruction and latent cycle-consistency losses, our model achieves a more stable training process while effectively learning the mapping between vector spaces.*

1. Introduction

The unsupervised alignment of embedding spaces is a fundamental challenge in machine learning, with the goal of mapping semantically related vectors from different domains or languages into a shared representation without access to paired data. This capability is crucial for tasks such as cross-lingual information retrieval, domain adaptation, and representation unification, where datasets often vary in modality, scale, or language.

Recent state-of-the-art methods, including Vec2Vec[Jha et al. 2025], have successfully adapted adversarial training and cycle-consistency from the image domain (e.g., CycleGAN) to align embedding spaces. However, these adversarial models are frequently difficult to train, prone to instability and mode collapse, and highly sensitive to hyperparameter tuning.

This project introduces a **Shared Autoencoder (SAE)** architecture as a stable, interpretable, and modular alternative to adversarial methods. Our primary contribution is a non-adversarial framework that dispenses with discriminators and instead relies on a combination of reconstruction, cycle-consistency, and distribution-matching losses to learn the translation. This approach is designed to preserve latent geometry while ensuring a more stable and reliable training process.

2. Related work

The challenge of unpaired translation was significantly advanced by the introduction of CycleGAN[Zhu et al. 2020], which demonstrated how cycle-consistency loss could be used in an adversarial framework to perform image-to-image translation without corresponding pairs. This paradigm was subsequently adapted for vector spaces. Notably,

Vec2Vec applied this concept to align embedding spaces, becoming a prominent method in the field.

However, the reliance on Generative Adversarial Networks (GANs) is a known source of difficulty. As detailed in surveys on the topic, GANs are often hard to train, suffer from issues like mode collapse, and require careful tuning of hyperparameters to converge. This inherent instability motivates the search for non-adversarial alternatives that can achieve similar results with greater reliability. Our work moves in this direction by replacing the adversarial component with robust reconstruction-based objectives.

3. Theoretical Background

Let two embeddings spaces be:

$$A = \{x_i \in R^{d_s}\}, \quad B = \{y_j \in R^{d_t}\}.$$

Our goal is to learn mappings into a shared latent space Z such that semantically aligned pairs occupy similar regions:

$$E_s : A \rightarrow Z, \quad D_s : Z \rightarrow A, \quad E_t : B \rightarrow Z, \quad D_t : Z \rightarrow B.$$

This is achieved without paired supervision via the following mechanisms:

3.1. Reconstruction and Cycle Consistency

$$\mathcal{L}_{rec} = \|x - D_s(E_s(x))\|^2 + \|y - D_t(E_t(y))\|^2$$

$$\mathcal{L}_{cyc} = \|E_t(D_t(E_s(x))) - E_s(x)\|^2 + \|E_s(D_s(E_t(y))) - E_t(y)\|^2$$

3.2. Distribution Matching

Instead of adversarial training, we use Sinkhorn Optimal Transport (OT) to match distributions:

$$\mathcal{L}_{dist} = \text{Sinkhorn}(D_t(E_s(x)), y) + \text{Sinkhorn}(D_s(E_t(y)), x)$$

This enforces statistical similarity between reconstructed and real embeddings.

3.3. Stability Regularization (VICReg)

To avoid representational collapse and maintain diversity:

$$\mathcal{L}_{stab} = \lambda_{sim} \|z_s - z_t\|^2 + \lambda_{var} \sum_i \text{ReLU}(1 - \sigma_i(z)) + \lambda_{cov} \|\text{offdiag}(\text{Cov}(z))\|^2$$

3.4. Geometry Preservation

We include local structure regularization to preserve neighborhood topology:

$$\mathcal{L}_{geo} = \mathcal{L}_{triplet} + \mathcal{L}_{laplacian}$$

ensuring that nearby points in A remain neighbors after translation into B .

4. Methodology

4.1. Model Design

- Shared Autoencoder with domain-specific encoders and decoders.
- Residual MLPs with GELU activations and BatchNorm.
- Latent dimension $d_z = 512$.
- LayerNorm applied to latent space for normalization.
- Implemented in PyTorch using the Vec2Vec training pipeline.

4.2. Experimental Setup

- Datasets:
 - Natural Questions (NQ) — open-domain QA dataset.
 - Enron Emails (ENRON) — business domain corpus.
- Embeddings:
 - CLIP
 - GTE
 - GTR
- Evaluation metrics:
 - Cosine similarity, Recall@k, Mean Rank.
 - Trustworthiness, Continuity, BLEU, ROUGE (optional text metrics).
 - Sinkhorn divergence for distributional alignment.
 - UMAP/t-SNE visualization of latent space.

References

- Jha, R., Zhang, C., Shmatikov, V., and Morris, J. X. (2025). Harnessing the universal geometry of embeddings.
- Saxena, D. and Cao, J. (2023). Generative adversarial networks (gans survey): Challenges, solutions, and future directions.
- Zhu, J.-Y., Park, T., Isola, P., and Efros, A. A. (2020). Unpaired image-to-image translation using cycle-consistent adversarial networks.